

Chapter 3

Basic Statistics

A data set is a collection of numerical observations. Before drawing conclusions from data, an observer needs concise descriptions of two properties: where the values tend to sit, and how much they vary around that location.

The first property is called the center of the data. The second is called the spread. This chapter defines three measures of center, two measures of spread, and examines the relationship between them. The choice of measure is not arbitrary: different measures reveal different features of the data, and the appropriate choice depends on the structure of the data set itself.

3.1 Measures of Center

A **measure of center** is a single number that summarizes the location of a data set on the number line. Three measures are in common use: the mean, the median, and the mode.

The Mean

Definition. *Mean.* The **mean** of a data set is the sum of all values divided by the number of values.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

where n is the number of values and $\bar{x}$ is read “ x -bar.”

The mean is the most commonly used measure of center. It uses every value in the data set.

Example 3.1.1. The high temperatures (in degrees Fahrenheit) recorded on seven consecutive days were:

58, 63, 61, 67, 70, 65, 62.

$$\bar{x} = \frac{58 + 63 + 61 + 67 + 70 + 65 + 62}{7} = \frac{446}{7} \approx 63.7.$$

The mean high temperature for the week was approximately 63.7°F.

The Median

Definition. Median. The **median** of a data set is the middle value when the values are arranged in order from smallest to largest. When the number of values is even, the median is the mean of the two middle values.

The median divides the data set in half: exactly half the values fall below the median and half fall above it.

Example 3.1.2. Find the median of the temperature data from Example 3.1.1.

Arranging in order:

58, 61, 62, **63**, 65, 67, 70.

With seven values, the middle value is in position 4. The median is **63**°F.

Example 3.1.3. (*Even number of values*) Find the median of the following eight quiz scores:

72, 85, 91, 78, 88, 65, 90, 80.

Arranging in order:

65, 72, 78, **80**, **85**, 88, 90, 91.

The two middle values (positions 4 and 5) are 80 and 85.

$$\text{Median} = \frac{80 + 85}{2} = \frac{165}{2} = 82.5.$$

The Mode

Definition. Mode. The **mode** of a data set is the value that appears most frequently. A data set may have one mode, more than one mode, or no mode if every value appears exactly once.

Example 3.1.4. Find the mode of each data set.

- (a) 3, 5, 5, 7, 8, 9, 5: the value 5 appears three times. The mode is 5.
- (b) 2, 4, 4, 6, 6, 8: both 4 and 6 appear twice. This data set has two modes: 4 and 6.
- (c) 10, 20, 30, 40: every value appears once. There is no mode.

Choosing a Measure of Center

The mean uses every value in its calculation. A single extreme value pulls the mean toward it without affecting the median. When a data set contains extreme values, the median is often a more representative description of center.

Example 3.1.5. (*Effect of an extreme value*) Six employees at a small company earn the following annual salaries (in thousands of dollars):

38, 42, 45, 47, 50, 310.

$$\text{Mean} = \frac{38 + 42 + 45 + 47 + 50 + 310}{6} = \frac{532}{6} \approx \$88,700.$$

Ordered data: 38, 42, 45, 47, 50, 310. The two middle values are 45 and 47.

$$\text{Median} = \frac{45 + 47}{2} = \$46,000.$$

The mean of \$88,700 is pulled upward by the one salary of \$310,000. The median of \$46,000 reflects what a typical employee earns. Five of the six employees earn less than the mean.

Exercises 3.1

3.1.1. Find the mean, median, and mode of each data set.

(a) 4, 7, 7, 9, 11, 13, 7, 6

(b) 22, 18, 30, 22, 25, 18, 29, 22, 31

(c) 0.5, 1.2, 0.8, 1.5, 0.9, 1.1

3.1.2. The following data records the number of customer complaints received by a store over ten days:

3, 0, 5, 2, 1, 4, 2, 8, 2, 3.

(a) Find the mean, median, and mode.

(b) On day 8, a particularly difficult situation led to 8 complaints. Describe the effect of removing this value from the data set on each of the three measures.

3.1.3. A real estate agent reports that the mean home price in a neighborhood is \$420,000, but the median is \$285,000. In two complete sentences, explain what this difference tells a potential buyer about the distribution of home prices in the neighborhood.

3.1.4. Seven students took a test. Six of them scored 72, 78, 84, 80, 76, 88. The mean score for all seven students is 80. What did the seventh student score?

Show your work.

3.2 Measures of Spread: Range

Two data sets can have the same mean but very different distributions of values around that mean. A measure of spread quantifies how much the values vary.

Definition. *Range.* The **range** of a data set is the difference between the maximum and minimum values.

$$\text{Range} = \text{maximum} - \text{minimum}$$

The range is easy to compute and gives a quick sense of the full extent of the data. It uses only two values and is sensitive to extreme observations.

Example 3.2.1. Two groups of students each took the same exam.

Group A scores: 62, 65, 68, 70, 72, 74, 78.

$$\text{Mean}_A = 69.9, \quad \text{Range}_A = 78 - 62 = 16.$$

Group B scores: 45, 58, 65, 72, 75, 80, 95.

$$\text{Mean}_B = 70.0, \quad \text{Range}_B = 95 - 45 = 50.$$

The two groups have nearly identical means but very different spreads. Group A's scores are clustered close to the center. Group B's scores are spread widely.

Exercises 3.2

3.2.1. Find the range of each data set.

(a) 14, 22, 9, 31, 18, 27

(b) -5, 0, 3, -2, 8, -7, 1

(c) 0.4, 1.1, 0.7, 1.8, 0.3

3.2.2. Two delivery drivers recorded the following number of deliveries per day over five days.

	Day 1	Day 2	Day 3	Day 4	Day 5
Driver A	12	14	11	13	15
Driver B	8	18	10	16	13

- (a) Compute the mean and range for each driver.
- (b) Which driver is more consistent? Justify your answer using the range.

3.2.3. A data set has a range of 40 and a minimum value of 15. What is the maximum value?

3.3 Measures of Spread: Sample Standard Deviation

The range measures spread using only the two most extreme values. The **sample standard deviation** uses every value in the data set. It measures the typical distance of each value from the mean.

Definition. *Sample standard deviation.* The **sample standard deviation** of a data set is:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

where $\bar{x}$ is the sample mean and n is the number of values in the sample.

Each term $(x_i - \bar{x})$ is the **deviation** of the value x_i from the mean. Squaring the deviations makes all terms non-negative. Taking the square root at the end returns the result to the same units as the original data.

Computing the sample standard deviation.

1. Find the mean $\bar{x}$.
2. Subtract the mean from each value to get the deviations $x_i - \bar{x}$.
3. Square each deviation.
4. Add the squared deviations and divide by $n - 1$.
5. Take the square root.

Example 3.3.1. Compute the sample standard deviation of the data set: 2, 4, 4, 6.

Step 1.

$$\bar{x} = \frac{2 + 4 + 4 + 6}{4} = \frac{16}{4} = 4.$$

Steps 2 and 3.

Value x_i	Deviation $x_i - \bar{x}$	Squared deviation $(x_i - \bar{x})^2$
2	$2 - 4 = -2$	4
4	$4 - 4 = 0$	0
4	$4 - 4 = 0$	0
6	$6 - 4 = 2$	4
Sum of squared deviations		8

Steps 4 and 5.

$$s = \sqrt{\frac{8}{3}} \approx 1.63.$$

The sample standard deviation of this data set is about 1.63 units.

Example 3.3.2. Compute the sample standard deviation of: 1, 3, 5, 7.

$$\bar{x} = \frac{1 + 3 + 5 + 7}{4} = \frac{16}{4} = 4.$$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
1	-3	9
3	-1	1
5	1	1
7	3	9
Sum =		20

$$s = \sqrt{\frac{20}{3}} \approx 2.58.$$

Comparing with Example 3.3.1: both data sets have mean 4, but this one has a larger sample standard deviation because the values are more spread out.

Example 3.3.3. (*Sample standard deviation of zero*) If every value in a data set is the same, every deviation is zero, and the sample standard deviation is zero.

Data: 5, 5, 5, 5.

$$\bar{x} = 5, \quad \text{all deviations} = 0, \quad s = 0.$$

A sample standard deviation of zero means there is no variability in the data.

Exercises 3.3

3.3.1. Compute the sample standard deviation of each data set. Show all steps using the deviation table.

(a) 3, 5, 5, 7

(b) 10, 20, 30, 40

(c) 6, 6, 9, 9

3.3.2. Without computing, decide which data set has a larger sample standard deviation. Explain your reasoning.

(a) Set A: 50, 51, 50, 49, 50 Set B: 30, 40, 50, 60, 70

(b) Set C: 2, 2, 2, 2, 2 Set D: 1, 2, 3, 4, 5

3.3.3. A data set has five values with mean $\bar{x} = 10$. The squared deviations are 4, 1, 0, 1, 4. Compute the sample standard deviation without knowing the original values.

3.3.4. Two machines fill bottles with juice. Machine A fills to 250, 251, 249, 250, 250 mL. Machine B fills to 245, 255, 248, 253, 249 mL.

(a) Compute the mean and sample standard deviation for each machine.

(b) Which machine is more consistent? Justify using the sample standard deviation.

3.4 Center and Spread Together

A complete description of a data set gives both a measure of center and a measure of spread. The two are used in pairs.

Measure of center	Companion measure of spread
Mean	Sample standard deviation
Median	Range

The mean and sample standard deviation are appropriate when the data is approximately symmetric with no extreme values. The median and range are more informative when the data contains extreme values or is heavily skewed.

Example 3.4.1. The following data records the ages of participants in two different programs.

Program A (yoga class):

28, 31, 34, 29, 33, 30, 32, 35.

$$\bar{x}_A = 31.5, \quad s_A \approx 2.45.$$

Program B (community center drop-in):

8, 15, 22, 45, 67, 72, 78, 83.

$$\bar{x}_B = 48.75, \quad \text{Median}_B = 56, \quad \text{Range}_B = 75.$$

Program A has a narrow age range and a small standard deviation. The mean and sample standard deviation describe it well. Program B spans a wide age range with no clear cluster. The median and range are more informative for Program B.

Exercises 3.4

3.4.1. For each data set, compute all five statistics: mean, median, mode, range, and sample standard deviation. Then state which pair (mean/SD or median/range) better describes the center and spread, and explain why.

(a) 12, 14, 14, 15, 16, 17, 14

(b) 5, 6, 7, 8, 9, 10, 85

3.4.2. A meteorologist records daily rainfall (in inches) for two weeks:

0, 0, 0.2, 0, 1.4, 0, 0, 0.1, 0, 0, 2.1, 0, 0, 0.

(a) Find the mean, median, and mode.

(b) Which measure of center best represents a typical day? Justify in two sentences.

3.4.3. A dataset has mean 20 and sample standard deviation 3. Another has mean 20 and sample standard deviation 10.

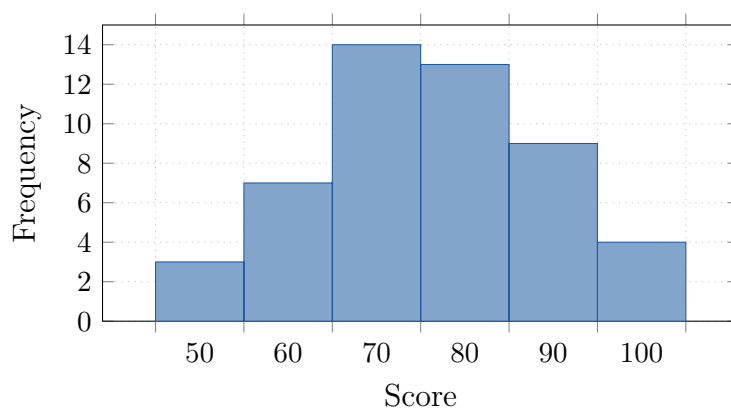
(a) Sketch a rough histogram for each, placing the center at 20.

(b) What does a larger sample standard deviation indicate about the distribution of values?

3.5 Visualizing Center and Spread

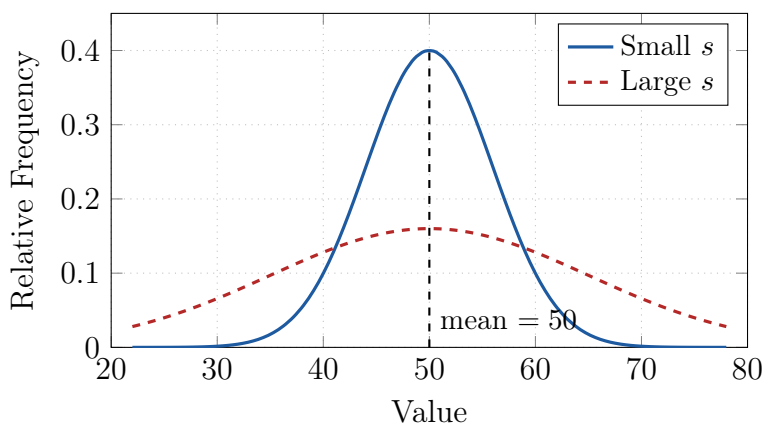
The histogram is the primary visual tool for understanding the distribution of a data set. Center and spread are visible in the shape of a histogram: the center is where the distribution is tallest, and the spread is how wide the distribution is.

Example 3.5.1. The following histogram shows the distribution of 50 exam scores.



The distribution is approximately symmetric and centered near 75–80. Most scores fall between 60 and 90. A small number of scores appear below 60 or above 90.

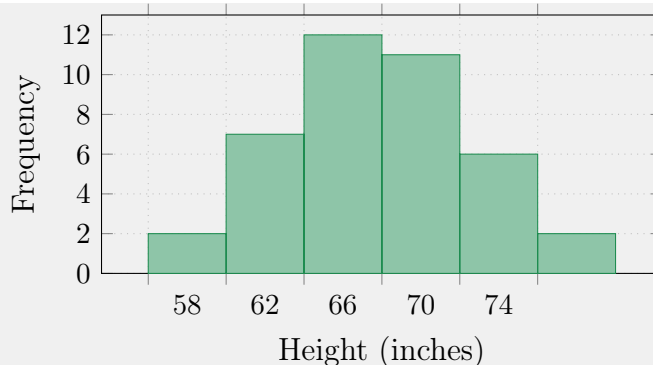
Example 3.5.2. (*Two distributions with the same mean*) The two distributions below both have mean 50, but different sample standard deviations.



The narrow distribution has a small sample standard deviation: values cluster tightly around the mean. The wide distribution has a large sample standard deviation: values are spread broadly around the mean. Both are centered at the same location.

Exercises 3.5

3.5.1. The histogram below shows the distribution of heights (in inches) for a sample of 40 adults.



- In which interval do most heights fall?
- Is the distribution symmetric or skewed?
- Would you expect the mean to be close to the median? Explain in one sentence.

3.5.2. Sketch two histograms with the same center but different spreads. Label the center on both. Below each sketch, write one sentence describing what each distribution tells an observer about the data.

3.5.3. A data set has mean 30 and sample standard deviation 2. A second data set has mean 30 and sample standard deviation 8. An observation of 36 is taken.

- How many sample standard deviations above the mean is 36 in each data set?
- In which data set is the value 36 more unusual? Explain in one sentence.

Chapter 3 Review

- The following data records the number of hours per week a sample of 10 students spend on homework:

4, 7, 5, 8, 6, 5, 9, 4, 6, 6.

- Find the mean, median, mode, and range.
 - Construct the deviation table and compute the sample standard deviation.
 - Describe the center and spread of this data set in two complete sentences.
- A company tracks the number of sick days taken by employees in one year. The data for eight employees is:

0, 1, 2, 1, 0, 3, 1, 18.

- (a) Compute the mean and median.
 - (b) Which measure better represents a typical employee? Explain.
 - (c) If the value 18 is removed, how do the mean and median each change? Compute both.
3. Three students each score a mean of 80 on five exams. Student A's scores are 78, 79, 80, 81, 82. Student B's scores are 60, 70, 80, 90, 100. Without computing, predict which student has the larger sample standard deviation. Then compute both sample standard deviations to verify.
4. A data set has six values. Five of them are 12, 15, 14, 13, 16. The mean of all six values is 14. Find the sixth value.
5. A histogram of test scores is skewed left.
- (a) What does it mean for a histogram to be skewed left?
 - (b) Would you expect the mean to be greater than, equal to, or less than the median? Explain in two sentences.
6. The temperatures recorded over five days in two cities are:

	Day 1	Day 2	Day 3	Day 4	Day 5
City A	68	70	69	71	67
City B	55	62	74	79	85

- (a) Compute the mean for each city.
 - (b) Compute the range and sample standard deviation for each city.
 - (c) Which city has more consistent temperatures? Justify using both spread measures.
7. A quality control engineer samples 6 ball bearings and measures their diameters (in mm):
- 9.98, 10.01, 10.00, 9.99, 10.02, 10.00.
- (a) Compute the mean and sample standard deviation.
 - (b) The specification requires that all bearings fall within 0.03 mm of the target diameter of 10.00 mm. Do all sampled bearings meet this requirement?